

Quiz 3: Conditional Expectation, Entropy, Stochastic Processes.

Solutions

Homework for lectures 5 and 6. Notation as in the lectures. Twelve problems, numbered 1–12 across the three parts, four per part; each needs the concept of its part.

Part A: Conditional Expectation

Notation as in the Conditional Expectation lecture: two dice Y (first) and Z (second), $X = Y + Z$ and $D = Y - Z$; the rows of the two-dice table are the events $\{Y = y\}$.

1. A supplier's defect rate is unknown: model it as Θ uniform on $[0, \frac{1}{5}]$. Given $\Theta = \theta$, the number D of defective items in a sample of 10 is binomial: $\mathbb{E}[D \mid \Theta = \theta] = 10\theta$ and $\mathbb{E}[D^2 \mid \Theta = \theta] = 10\theta(1 - \theta) + 100\theta^2$. Compute $\mathbb{E}[D]$ and $\text{Var}(D)$, and compare $\text{Var}(D)$ with the variance $10 \cdot 0.1 \cdot 0.9$ of a binomial with the average defect rate 0.1. Then compute $\mathbb{P}(D = 0)$ directly, by integrating over θ , and compare with 0.9^{10} .

Solution. $\mathbb{E}[D] = 10\mathbb{E}[\Theta] = 1$. $\mathbb{E}[D^2] = 10\mathbb{E}[\Theta] + 90\mathbb{E}[\Theta^2] = 1 + 90 \cdot \frac{1}{75} = \frac{11}{5}$ (as $\mathbb{E}[\Theta^2] = \int_0^{1/5} 5\theta^2 d\theta = \frac{1}{75}$), so $\text{Var}(D) = \frac{11}{5} - 1 = \frac{6}{5} = 1.2 > 0.9$; *uncertainty about the rate increases the variance of D . Directly, $\mathbb{P}(D = 0) = \int_0^{1/5} 5(1 - \theta)^{10} d\theta = \frac{5}{11}(1 - 0.8^{11}) \approx 0.416$, against $0.9^{10} \approx 0.349$. Each further value $\mathbb{P}(D = d)$ is another such integral, and computing $\mathbb{E}[D]$ from the pmf would require all eleven.*

2. Toss a fair coin until the pattern THH first appears; then the same for THT and for TTT. Compute the three expected waiting times by conditioning on the first toss from each of the three states “no progress”, “T”, “TH” (respectively “TT”). Which pattern has the smallest expected waiting time, and why do three patterns of the same length have different expected waiting times?

Solution. Write m_0, m_1, m_2 for the expected number of further tosses from the states “no progress”, “T”, and “T then the second letter”. Conditioning on the next toss: THH: $m_0 = 1 + \frac{1}{2}(m_0 + m_1)$, $m_1 = 1 + \frac{1}{2}(m_1 + m_2)$, $m_2 = 1 + \frac{1}{2}(0 + m_1)$, so $(m_0, m_1, m_2) = (8, 6, 4)$. THT: the same first two equations, but $m_2 = 1 + \frac{1}{2}(0 + m_0)$, so $(10, 8, 6)$. TTT: $m_0 = 1 + \frac{1}{2}(m_0 + m_1)$, $m_1 = 1 + \frac{1}{2}(m_0 + m_2)$, $m_2 = 1 + \frac{1}{2}(0 + m_0)$, so $(14, 12, 8)$. So THH takes 8 tosses on average, THT 10, TTT 14. The difference lies in what remains after a failed attempt: after TH, the wrong toss for THH is a T, which begins a new attempt; the wrong toss for THT is an H, which contributes nothing; and for TTT a failing H at any stage discards all progress.

3. Two dice, $D = Y - Z$, $X = Y + Z$. (a) Show that $\mathbb{E}[D \mid X] = 0 = \mathbb{E}[D]$ as random variables, by considering how the diagonal $\{X = x\}$ lies relative to the main diagonal. (b) Show that $\mathbb{E}[X \mid D] = 7 = \mathbb{E}[X]$ (which cells form $\{D = d\}$, and what is X on them?). Deduce $\text{Cov}(X, D) = 0$. (c) Show that X and D are nevertheless *not* independent, by comparing the parities of X and D or by computing $\mathbb{P}(D = 5, X = 7)$ against $\mathbb{P}(D = 5)\mathbb{P}(X = 7)$. Which result of the lecture does this illustrate?

Solution. (a) If the cell $(y, x - y)$ lies on the diagonal $\{X = x\}$ then so does $(x - y, y)$, and the two have opposite values of D ; so the average of D over each diagonal is 0, and $\mathbb{E}[D \mid X] = 0$, a constant. (b) $\{D = d\}$ is the diagonal of cells $(y, y - d)$ with both coordinates in $\{1, \dots, 6\}$; on it $X = 2y - d$ and y runs over an interval symmetric about $\frac{7+d}{2}$, so the average of X is $2 \cdot \frac{7+d}{2} - d = 7$. Hence $\mathbb{E}[X \mid D] = 7$, and $\text{Cov}(X, D) = \mathbb{E}[D \mathbb{E}[X \mid D]] - \mathbb{E}[X]\mathbb{E}[D] = 7\mathbb{E}[D] - 7\mathbb{E}[D] = 0$. (c) Not independent: $X = D + 2Z$, so X and D always have the same parity, and $\mathbb{P}(D \text{ odd}, X \text{ even}) = 0 \neq \mathbb{P}(D \text{ odd})\mathbb{P}(X \text{ even})$. Or: $\mathbb{P}(D = 5, X = 7) = \mathbb{P}(Y = 6, Z = 1) = \frac{1}{36}$, while $\mathbb{P}(D = 5)\mathbb{P}(X = 7) = \frac{1}{36} \cdot \frac{6}{36}$. This illustrates the distinction between equal row averages and equal row distributions: $\mathbb{E}[X \mid D] = \mathbb{E}[X]$ says that every diagonal has average 7, whereas independence would require every diagonal to carry the distribution of X , which a diagonal of 3 cells cannot.

4. Let $X_1, X_2, \dots$ be independent, each uniform on $\{1, \dots, 6\}$, $\mu = \mathbb{E}[X_i] = 3.5$, and let N be the index of the first roll showing 5 or 6; N is determined by the X_i and not independent of them. (a) Let $S = X_1 + \dots + X_N$ include the stopping roll. Compute $\mathbb{E}[S]$ by conditioning on the first roll, and compare with $\mu \mathbb{E}[N]$. (b) Let $S' = X_1 + \dots + X_{N-1}$ exclude it. Compute $\mathbb{E}[S']$ the same way and compare with $\mu \mathbb{E}[N - 1]$. Explain the two comparisons.

Solution. N is geometric with success probability $\frac{1}{3}$: $\mathbb{E}[N] = 3$, $\mu \mathbb{E}[N] = 10.5$, $\mu \mathbb{E}[N - 1] = 7$. (a) Condition on $X_1 = k$. If $k \in \{5, 6\}$: $S = k$. If $k \leq 4$: $S = k + S''$ with S'' distributed as S . So $\mathbb{E}[S] = \frac{1}{6}(5 + 6) + \frac{1}{6} \sum_{k=1}^4 (k + \mathbb{E}[S]) = \frac{11}{6} + \frac{10}{6} + \frac{4}{6} \mathbb{E}[S]$, giving $\mathbb{E}[S] = \frac{21}{2} = 10.5 = \mu \mathbb{E}[N]$. (b) Now $S' = 0$ if $k \in \{5, 6\}$ and $S' = k + S'''$ otherwise, with S''' distributed as S' : $\mathbb{E}[S'] = \frac{10}{6} + \frac{4}{6} \mathbb{E}[S']$, so $\mathbb{E}[S'] = 5 \neq 7$. The random-sums theorem applies to neither sum, since N depends on the X_i . In (a) the decision to stop after n rolls depends only on the first n rolls, and the formula still holds (Wald's identity). In (b) the last roll is excluded because it is large: $\mathbb{E}[S'] = \mathbb{E}[S] - \mathbb{E}[X_N] = 10.5 - 5.5 = 5$, and the excluded term has mean 5.5, not μ . The formula treats the $N - 1$ terms as typical rolls, whereas they are the rolls that showed at most four.

Part B: Entropy

Logarithms are base 2, the unit is the bit, and $0 \log 0 = 0$. Notation as in the Entropy lecture: p_X is the pmf of X ; $H(X)$ is entropy and $h(p)$ the binary entropy function; $L(c)$ is the average length of the prefix-free code c and L^* the optimal average length; $H(P, Q)$ is the cross-entropy of the truth P with the code or model Q , $D(P \parallel Q) = H(P, Q) - H(P)$; $H(X \mid Y)$ and $I(X; Y)$ are conditional entropy and mutual information.

5. Optimality of the rover's code. Let $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$. (a) Using Kraft's inequality and Gibbs' lemma, show that every prefix-free code for four symbols has $L(c) \geq 1.75$ under p , so 0, 10, 110, 111 is optimal. (b) Show that the only length profiles for four symbols with Kraft sum exactly 1 are (2, 2, 2, 2) and (1, 2, 3, 3), up to order. (Consider the shortest length.) (c) Take as given that an optimal code has Kraft sum 1 (otherwise a codeword can be shortened). Deduce that for every pmf on four symbols the optimal code is either the fixed-length code or 0, 10, 110, 111 with the symbols assigned in decreasing order of probability, whichever has the smaller L . (d) Apply (c) to $p = (0.45, 0.25, 0.15, 0.15)$ and to $p = (0.3, 0.3, 0.2, 0.2)$, and compare L^* with H in each case. Why can the source coding theorem alone not decide which code is optimal here?

Solution. (a) Put $a(x) = 2^{-\ell_c(x)}$; Kraft gives $\sum a \leq 1$ and $\ell_c(x) = \log \frac{1}{a(x)}$. Gibbs' lemma: $L(c) = \sum_x p(x) \log \frac{1}{a(x)} \geq \sum_x p(x) \log \frac{1}{p(x)} = H(p) = 1.75$, with equality iff $a = p$, i.e. lengths (1, 2, 3, 3). (b) Order $\ell_1 \leq \ell_2 \leq \ell_3 \leq \ell_4$. If $\ell_1 \geq 2$ then $\sum 2^{-\ell_i} \leq 4 \cdot \frac{1}{4} = 1$ with equality only for (2, 2, 2, 2). If $\ell_1 = 1$, the other three sum to $\frac{1}{2}$; $\ell_2 = 1$ is impossible (two

words of length 1 already fill the Kraft sum and leave nothing for ℓ_3, ℓ_4), and if $\ell_2 \geq 3$ they sum to at most $\frac{3}{8}$, so $\ell_2 = 2$ and $2^{-\ell_3} + 2^{-\ell_4} = \frac{1}{4}$ with $\ell_3, \ell_4 \geq 3$, forcing $(3, 3)$. Hence $(1, 2, 3, 3)$. (c) An optimal code has one of the two profiles of (b). For $(1, 2, 3, 3)$, $L = \ell_1 p_{(1)} + \ell_2 p_{(2)} + 3p_{(3)} + 3p_{(4)}$ is smallest when the shortest words go to the most probable symbols (swapping two assignments that violate this can only lower L). So $L^* = \min(2, p_{(1)} + 2p_{(2)} + 3p_{(3)} + 3p_{(4)})$ with $p_{(1)} \geq \dots \geq p_{(4)}$. (d) $(0.45, 0.25, 0.15, 0.15)$: $L(1, 2, 3, 3) = 0.45 + 0.5 + 0.45 + 0.45 = 1.85 < 2$, so $0, 10, 110, 111$ is optimal, $L^* = 1.85$, while $H = 1.840$. $(0.3, 0.3, 0.2, 0.2)$: $L(1, 2, 3, 3) = 0.3 + 0.6 + 0.6 + 0.6 = 2.1 > 2$, so the fixed-length code is optimal, $L^* = 2$, $H = 1.971$. The theorem says only that $H \leq L^* < H + 1$, i.e. $L^* \in [1.84, 2.84)$ and $[1.97, 2.97)$; both candidates lie in each window, and the choice requires the enumeration in (b).

6. Coin A shows heads with probability 0.9, coin B is fair. One coin is chosen uniformly at random ($C \in \{A, B\}$) and tossed once, giving $X \in \{H, T\}$. (a) Compute the pmf of X and $H(X)$. (b) Compute $H(X | C = A)$, $H(X | C = B)$ and $H(X | C)$. (c) Compare $H(X)$ with $H(X | C)$ and state the property of h that this illustrates. What is $I(X; C)$? (d) Compute $\mathbb{P}(C = A | X = H)$ and $\mathbb{P}(C = A | X = T)$, then $H(C | X)$, and check $H(C) - H(C | X) = I(X; C)$.

Solution. (a) $\mathbb{P}(X = H) = \frac{1}{2} \cdot 0.9 + \frac{1}{2} \cdot 0.5 = 0.7$; $H(X) = h(0.7) = 0.881$ bits. (b) $h(0.9) = 0.469$, $h(0.5) = 1$, $H(X | C) = \frac{1}{2}(0.469 + 1) = 0.734$. (c) $H(X) = 0.881 > 0.734 = H(X | C)$: the entropy of a mixture exceeds the average of the entropies, which is the concavity of h : $h(\frac{0.9+0.5}{2}) \geq \frac{1}{2}h(0.9) + \frac{1}{2}h(0.5)$. The difference $I(X; C) = 0.147$ bits is the information one toss carries about which coin was chosen. (d) $\mathbb{P}(A | H) = \frac{0.45}{0.7} = \frac{9}{14} = 0.643$, $\mathbb{P}(A | T) = \frac{0.05}{0.3} = \frac{1}{6}$. $H(C | X) = 0.7 h(\frac{9}{14}) + 0.3 h(\frac{1}{6}) = 0.7 \cdot 0.940 + 0.3 \cdot 0.650 = 0.853$, and $H(C) - H(C | X) = 1 - 0.853 = 0.147 = I(X; C)$.

7. The rover has two missions, equally likely: $P_1 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ and $P_2 = (\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2})$ for (up, down, left, right). One code, built for a pmf Q , must be installed before the mission is known; its cost is $\frac{1}{2}H(P_1, Q) + \frac{1}{2}H(P_2, Q)$ bits per instruction. (a) Show that this cost equals $H(\bar{P}, Q)$ with $\bar{P} = \frac{1}{2}(P_1 + P_2)$, and compute $\bar{P}$. (b) Find the Q that minimises the cost and the minimal cost. Compare with keeping the old code ($Q = P_1$), with the fixed-length code, and with the 1.75 bits achievable if the mission were known. (c) Show that the average waste $\frac{1}{2}D(P_1 \| \bar{P}) + \frac{1}{2}D(P_2 \| \bar{P})$ equals $I(M; X)$, the mutual information between the mission $M \in \{1, 2\}$ and one instruction X , and compute it. (d) Codeword lengths must be integers. Using Problem 5(b), find the best single-symbol code for the two-mission problem. What must be done to approach the cost in (b)?

Solution. (a) $\frac{1}{2} \sum_x P_1(x) \log \frac{1}{Q(x)} + \frac{1}{2} \sum_x P_2(x) \log \frac{1}{Q(x)} = \sum_x \bar{P}(x) \log \frac{1}{Q(x)}$: cross-entropy is linear in its first argument. $\bar{P} = (\frac{5}{16}, \frac{3}{16}, \frac{3}{16}, \frac{5}{16})$. (b) By Gibbs, $H(\bar{P}, Q) \geq H(\bar{P})$ with equality iff $Q = \bar{P}$. Minimal cost $H(\bar{P}) = 2 \cdot \frac{5}{16} \log \frac{16}{5} + 2 \cdot \frac{3}{16} \log \frac{16}{3} = 1.049 + 0.906 = 1.954$ bits. Old code: $\frac{1}{2}(1.75 + 2.625) = 2.1875$. Fixed-length: 2. Known mission: 1.75. (c) The average waste is $H(\bar{P}, \bar{P}) - \frac{1}{2}H(P_1) - \frac{1}{2}H(P_2) = H(\bar{P}) - \frac{1}{2}(H(P_1) + H(P_2))$. With M uniform, X has pmf $\bar{P}$ and $H(X | M = m) = H(P_m)$, so this is $H(X) - H(X | M) = I(M; X) = 1.954 - 1.75 = 0.204$ bits, the information one instruction carries about the mission, which is also the average waste from not knowing it. (d) Profile $(2, 2, 2, 2)$: cost 2. Profile $(1, 2, 3, 3)$ with the short words on the two probable symbols: $\frac{5}{16} \cdot 1 + \frac{5}{16} \cdot 2 + \frac{3}{16} \cdot 3 + \frac{3}{16} \cdot 3 = \frac{33}{16} = 2.0625$. The fixed-length code is the better of the two. To approach 1.954, code blocks of instructions with lengths built for $\bar{P}$ (block coding theorem).

8. A binary text source is modelled as a Markov chain on $S = \{0, 1\}$ with transition matrix $Q = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$. (a) Find the stationary pmf π . (b) Compute the entropy rate

$H(X_{i+1} | X_i)$ and the monographic entropy $H(\pi)$, and identify their difference. (c) Compute $H(X_1, X_2, X_3)$ for the chain started from π , once by the chain rule and once by listing the eight probabilities. (d) Give the perplexity of the Markov model, of the monographic model and of the uniform model. (e) What does a compressor built on the monographic model pay per symbol on this text, and how much of it is waste?

Solution. (a) Balance $0 \rightarrow 1$ against $1 \rightarrow 0$: $0.1\pi_0 = 0.3\pi_1$, so $\pi = (\frac{3}{4}, \frac{1}{4})$. (b) Rate $= \frac{3}{4}h(0.1) + \frac{1}{4}h(0.3) = 0.75 \cdot 0.469 + 0.25 \cdot 0.881 = 0.572$ bits per symbol; $H(\pi) = h(\frac{1}{4}) = 0.811$. The difference $0.239 = I(X_i; X_{i+1})$ is the mutual information of consecutive symbols. (c) Chain rule with the Markov property: $H(\pi) + 2 \cdot 0.572 = 1.955$. Listing: $\pi(a)Q(a, b)Q(b, c)$ over the eight triples is 0.6075, 0.0675, 0.0225, 0.0525, 0.0675, 0.0075, 0.0525, 0.1225 (for 000, 001, 010, 011, 100, 101, 110, 111), whose entropy is 1.955. (d) $2^{0.572} = 1.49$; $2^{0.811} = 1.75$; $2^1 = 2$. (e) Its cost is $\sum_x \pi(x) \log \frac{1}{\pi(x)} = H(\pi) = 0.811$ bits per symbol, since the symbol frequencies are π ; the waste is $0.811 - 0.572 = 0.239$ bits per symbol, the mutual information of (b).

Part C: Stochastic Processes

Notation as in the Stochastic Processes lecture: B is a standard Brownian motion; $\mathbb{P}_i, \mathbb{E}_i$ refer to a start at i ; $p = 1 - q$ is the probability of winning a round, $r = q/p$. $\mathbf{P} = (p(i, j))$ is the transition matrix, row i the conditional pmf of the next state given the current state i ; π is a stationary distribution; $h_i = \mathbb{P}_i(\text{reach the target})$ and $k_i = \mathbb{E}_i[\text{time to absorption}]$ are hitting probabilities and times. Φ is the $N(0, 1)$ cdf; values used below: $\Phi(0.100) = 0.540$, $\Phi(0.500) = 0.691$, $\Phi(0.674) = 0.750$, $\Phi(1.000) = 0.841$.

9. The reflection principle for Brownian motion: $\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a)$. Let T_1 be the first time B reaches 1. (a) Compute $\mathbb{P}(\max_{s \leq 4} B_s \geq 2)$. (b) Write $\mathbb{P}(T_1 \leq t)$ in terms of Φ and evaluate it at $t = 1, 4, 100$. (c) Find the median of T_1 (the t with $\mathbb{P}(T_1 \leq t) = \frac{1}{2}$). (d) By scaling, what is the median of T_a ? (e) The lecture showed $\mathbb{E}[T_1] = \infty$. Reconcile this with a median of about 2.

Solution. (a) $2\mathbb{P}(B_4 \geq 2) = 2\mathbb{P}(N(0, 1) \geq 1) = 2(1 - \Phi(1)) = 0.317$. (b) $\{T_1 \leq t\} = \{\max_{s \leq t} B_s \geq 1\}$, so $\mathbb{P}(T_1 \leq t) = 2(1 - \Phi(1/\sqrt{t}))$: 0.317 at $t = 1$; $2(1 - \Phi(0.5)) = 0.617$ at $t = 4$; $2(1 - \Phi(0.1)) = 0.920$ at $t = 100$. (c) $2(1 - \Phi(1/\sqrt{t})) = \frac{1}{2}$ iff $\Phi(1/\sqrt{t}) = \frac{3}{4}$ iff $1/\sqrt{t} = 0.674$: median $t = 1/0.674^2 = 2.20$. (d) $\frac{1}{a}B_{a^2t}$ is a Brownian motion, and it reaches 1 exactly when B reaches a , at a time smaller by the factor a^2 ; so T_a has the distribution of a^2T_1 and median $2.20a^2$. (e) Half the paths reach 1 within 2.2 time units, but the tail is heavy: $\mathbb{P}(T_1 > t) = 2\Phi(1/\sqrt{t}) - 1 \approx \sqrt{2/(\pi t)}$, and $\int_0^\infty t^{-1/2} dt = \infty$. The median depends only on $\mathbb{P}(T_1 \leq t)$ for $t \leq 2.2$; the mean equals $\int_0^\infty \mathbb{P}(T_1 > t) dt$, which diverges.

10. Bold play. $p = 0.4$, fortune £3, target £10. (a) Timid play (£1 per round): compute h_3 and $\mathbb{E}_3[T]$ from the lecture's ruin formulas. (b) Bold play: each round stake $\min(i, 10 - i)$, so the fortune moves from 3 to 6 or 0, from 6 to 10 or 2, from 2 to 4 or 0, from 4 to 8 or 0, from 8 to 10 or 6. Draw the chain on $\{0, 2, 3, 4, 6, 8, 10\}$ and write the hitting system for $h_i = \mathbb{P}_i(\text{reach } 10)$. (c) Solve it and compare h_3 with (a). (d) Compute the expected number of bets under bold play from £3. (e) Repeat (c) for the fair game $p = \frac{1}{2}$. Why is the answer the same as for timid play there?

Solution. (a) $r = 1.5$: $h_3 = \frac{1.5^3 - 1}{1.5^{10} - 1} = 0.042$; $\mathbb{E}_3[T] = \frac{3}{0.2} - \frac{10}{0.2} \cdot \frac{1 - 1.5^3}{1 - 1.5^{10}} = 12.9$ rounds. (b) $h_0 = 0$, $h_{10} = 1$, $h_3 = ph_6$, $h_6 = p + qh_2$, $h_2 = ph_4$, $h_4 = ph_8$, $h_8 = p + qh_6$. (c) From $h_8 = p + qh_6$ and $h_6 = p + qh_2 = p + qp^2h_8$: $h_8 = p + qp + q^2p^2h_8$, so $h_8 = \frac{p(1+q)}{1-q^2p^2} = \frac{0.64}{0.9424} = \frac{400}{589} = 0.679$; then $h_6 = \frac{274}{589} = 0.465$, $h_4 = \frac{160}{589} = 0.272$, $h_2 = \frac{64}{589} = 0.109$, and

$h_3 = ph_6 = \frac{548}{2945} = 0.186$: about 4.4 times the timid-play value 0.042. (d) $k_0 = k_{10} = 0$, $k_3 = 1 + pk_6$, $k_6 = 1 + qk_2$, $k_2 = 1 + pk_4$, $k_4 = 1 + pk_8$, $k_8 = 1 + qk_6$: $k_3 = \frac{1073}{589} = 1.82$ bets, against 12.9 rounds of timid play. (e) With $p = \frac{1}{2}$ the system gives $h_3 = \frac{3}{10}$, the timid value. A fair game preserves expected fortune whatever the stakes, so $\mathbb{E}[X_T] = 3 = 10h_3$ for every strategy that stops at 0 or 10.

11. PageRank with damping $d = 0.85$. (a) Three pages: $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow A$. Write $\mathbf{H}$ and $\mathbf{P} = 0.85\mathbf{H} + 0.05\mathbf{J}$. (b) Solve $\pi\mathbf{P} = \pi$ and rank the pages. A and B have one in-link each; why does A rank higher? (c) Find π without damping ($d = 1$). (d) Add a page D with links $C \rightarrow D$ (so C now links to A and D) and $D \rightarrow D$ only. Find π without damping, and with $d = 0.85$. Describe the effect of D on the surfer's long-run behaviour, and the effect of damping on it.

Solution. (a) $\mathbf{H} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$, $\mathbf{P} = \begin{pmatrix} 0.05 & 0.475 & 0.475 \\ 0.05 & 0.05 & 0.9 \\ 0.9 & 0.05 & 0.05 \end{pmatrix}$. (b) $\pi\mathbf{P} = \pi$ with $\sum \pi = 1$:

$\pi_A = 0.05 + 0.85\pi_C$, $\pi_B = 0.05 + 0.425\pi_A$, $\pi_C = 0.05 + 0.425\pi_A + 0.85\pi_B$; solving, $\pi = (\frac{686}{1769}, \frac{380}{1769}, \frac{703}{1769}) = (0.388, 0.215, 0.397)$: $C > A > B$. A's one in-link is the only link of the top page C, so it carries all of C's weight; B's in-link is one of two from A and carries half of A's. (c) $\pi\mathbf{H} = \pi$: $\pi_A = \pi_C$, $\pi_B = \frac{1}{2}\pi_A$, $\pi_C = \frac{1}{2}\pi_A + \pi_B$: $\pi = (\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$. (d) Without damping, D is absorbing: once the surfer arrives (which happens with probability 1: every visit to C leads to D with probability $\frac{1}{2}$) it never leaves, and the only stationary distribution is $\pi = (0, 0, 0, 1)$. With $d = 0.85$, solving $\pi\mathbf{P} = \pi$ gives $\pi = (0.101, 0.080, 0.148, 0.671)$: D still holds two thirds of the mass, because the surfer stays on D until the next teleport, a geometric number of steps with mean $1/0.15 = 6.7$, and every teleport lands on D a quarter of the time. Damping keeps every page's rank positive but does not remove the effect of a self-linking trap; search engines treat such pages separately.

12. Tennis at deuce. From deuce the server wins each point with probability p , independently. States: D (deuce), I (advantage server), O (advantage receiver), W (game won), L (game lost). From D the next state is I or O; from I it is W or D; from O it is D or L. (a) Write the transition matrix. (b) Write the hitting system for $h_i = \mathbb{P}_i(\text{reach W})$ and solve it; show $h_D = \frac{p^2}{p^2 + q^2}$. (c) Write the system for $k_i = \mathbb{E}_i[\text{points until the game ends}]$ and show $k_D = \frac{2}{1 - 2pq}$. (d) Evaluate h_D and k_D for $p = 0.5, 0.55, 0.6, 0.7$. What does a 55/45 edge per point become per game? (e) Express h_D as an infinite series by summing over the number of points played, and explain why the first-step system avoids the series.

Solution. (a) Order D, I, O, W, L: $\mathbf{P} = \begin{pmatrix} 0 & p & q & 0 & 0 \\ q & 0 & 0 & p & 0 \\ p & 0 & 0 & 0 & q \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$. (b) $h_W = 1$, $h_L = 0$, $h_D = ph_I +$

qh_O , $h_I = p + qh_D$, $h_O = ph_D$. Substituting, $h_D = p^2 + pqh_D + pqh_D$, so $h_D(1 - 2pq) = p^2$ and, as $1 - 2pq = p^2 + q^2$, $h_D = \frac{p^2}{p^2 + q^2}$. For $p = 0.6$: $h_D = \frac{9}{13}$, $h_I = \frac{57}{65}$, $h_O = \frac{27}{65}$. (c) $k_W = k_L = 0$, $k_D = 1 + pk_I + qk_O$, $k_I = 1 + qk_D$, $k_O = 1 + pk_D$. Then $k_D = 1 + p + q + 2pqk_D = 2 + 2pqk_D$, so $k_D = \frac{2}{1 - 2pq}$; for $p = 0.6$, $k_D = \frac{50}{13} = 3.85$ points. (d) $h_D = 0.5, 0.599, 0.692, 0.845$ and $k_D = 4, 3.96, 3.85, 3.45$. A 55/45 point becomes a 60/40 game from deuce; the advantage is amplified because the game ends only when one player is two points ahead. (e) The game ends after $2m$ more points with probability $(2pq)^{m-1}(p^2 + q^2)$, and the server wins with probability $\sum_{m \geq 1} (2pq)^{m-1}p^2$, an infinite series. The first-step system uses the restart at deuce to write h_D in terms of itself, so the series is summed implicitly.